EVALUATING GRADIENT AGREEMENT
FOR LEARNING-RATE CONTROL

For a neural network trained on the MNIST handwritten digit dataset, does adjusting the stochastic gradient descent learning rate based on gradient agreement help it learn its weights faster or achieve higher accuracy?

5 matched seeds
Matched conditions within each seed

97.97%
Guided mean final validation accuracy

+28.7% time
Guided time to 97% vs fixed SGD

01 METHOD

Compare the direction of this batch's gradient with the previous batch, then choose one learning rate for the whole model.

Images > Predict > Error > Gradient

Compare directions > Choose rate > Update

Cosine similarity measures direction

Gray: previous gradient | Color: current gradient

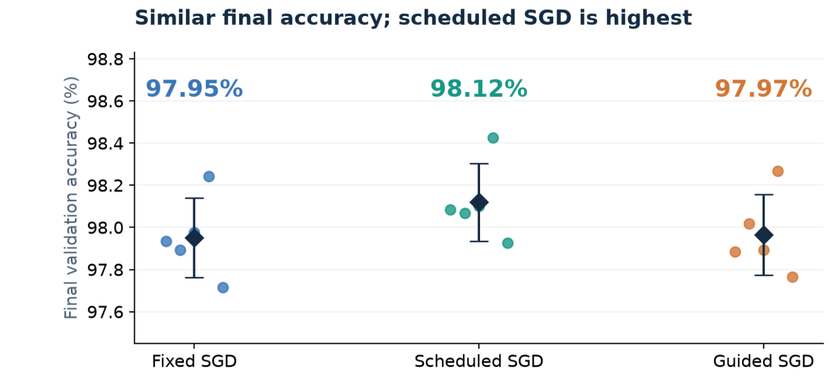


Strong agreement: s > 0.50	0.50
Weak agreement: 0 <= s <= 0.50	0.375
Opposing directions: s < 0	0.50

First update or undefined direction: 0.50.
The opposing-direction rate was changed to 0.50 during experimentation. Alignment does not mean a correct prediction.

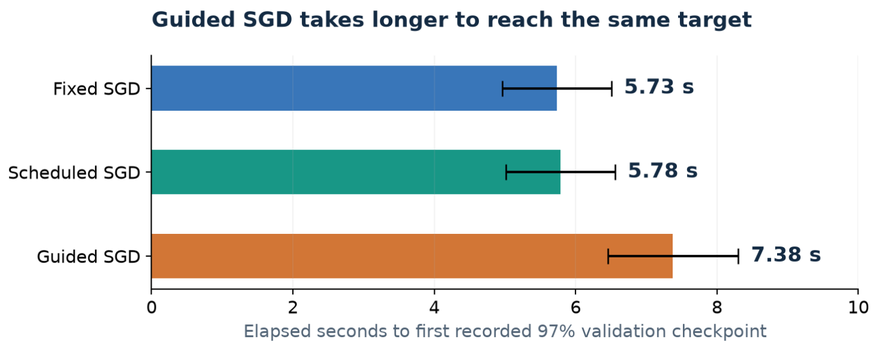
Baselines: Fixed LR 0.50 throughout.
Scheduled S13: 0.50 for epochs 1-7;
0.25 for epochs 8-10.

02 VALIDATION ACCURACY



Dots: five seeds. Diamonds: mean. Whiskers: sample standard deviation.

03 TIME TO TARGET



Finding: Guided SGD achieved similar validation accuracy to fixed SGD and required more elapsed time. All three methods reached 97% at the same epoch within each seed.

Setup: 784 > 256 > ReLU > 10 | 48,000 train / 12,000 validation | batch 64 | 10 epochs | seeds 42, 123, 456, 789, 2026

Exploratory validation evidence: test evaluation occurred before the final five-seed comparison. Times are measured at validation checkpoints. Dots show seeds; whiskers show sample SD. See accompanying article for sources and limitations.